Ejercicio #3

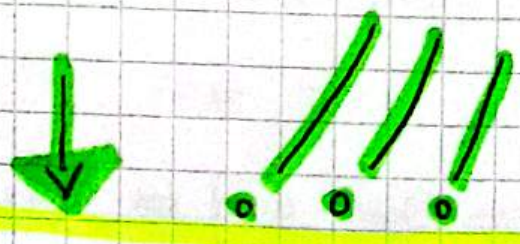
Sistemas de numeración

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1) Convertir de decimal a binario

1a) 44 1b) 60

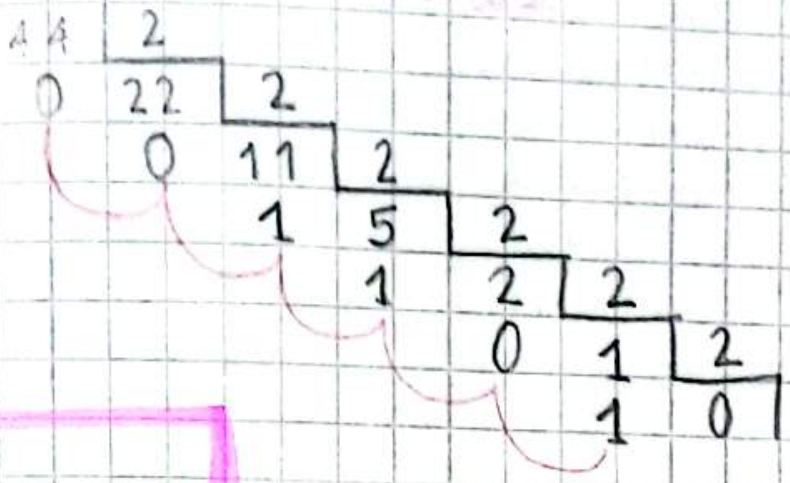
Solución



NOTA = Los digitos de 3 los divido de a dos cifras. Do el resulto correc.

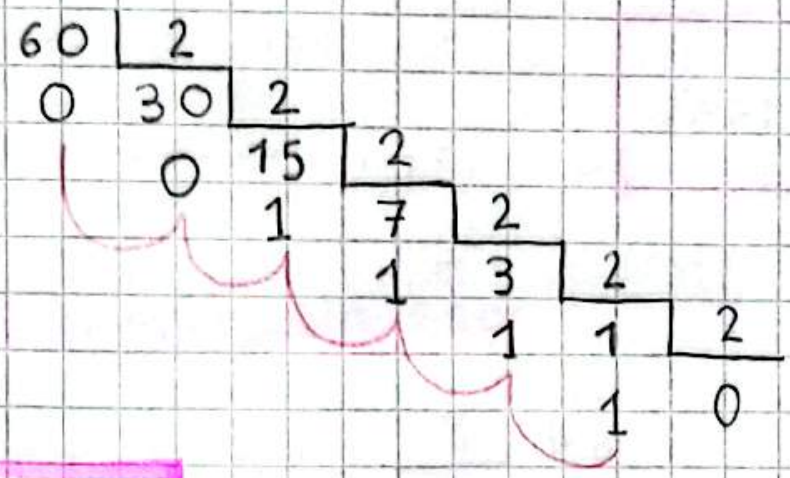
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1a) 44



$44_{10} = 101100_2$

1b) 60



$60_{10} = 111100_2$

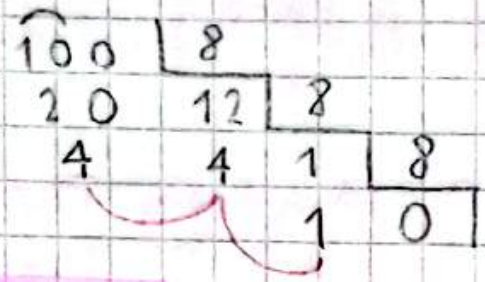
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2) Convertir de decimal a octal:

2a) 100 2b) 64

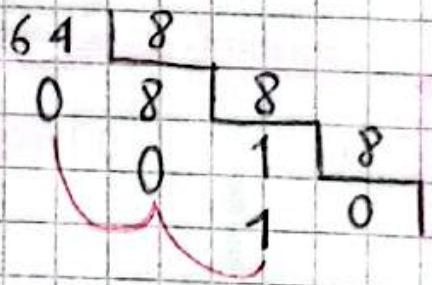
Solución

2a) 100



$100_{10} = 144_8$

2b) 64



$64_{10} = 100_8$

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3) Convertir de decimal a hexadecimal

3a) 500 3b) 240

Solución

3a) 500

Nota en el sistema hexadecimal =

A = (10)
B = (11)
C = (12)
D = (13)
E = (14)
F = (15)

ahora sí Solución 3)a 500

500	16		
20	31	16	
4	15	1	16
		1	0

$500_{10} = 1F4_{16}$

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3b) 240

240	16	
80	15	16
0	15	0

$240_{10} = F0_{16}$

4) Convertir a base decimal

4a) 110101 (binario) 4b) 634 (octal)

4c) AAA (decimal) 4d) 1000001 (Binario)

4e) 3765 (Octa)) 4f) F000C (hexadecimal)

4g) 11111 (binario) 4h) 11111 (octal)

4i) 1111 (hexadecimal)

Solución

4a) 110101 (binario)

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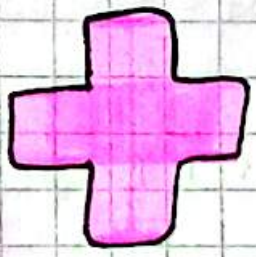
1	1	0	1	0	1
2^5	2^4	2^3	2^2	2^1	2^0
32	16	8	4	2	1

$1 \times 32 = 32$
$1 \times 16 = 16$
$0 \times 8 = 0$
$1 \times 4 = 4$
$0 \times 2 = 0$
$1 \times 1 = 1$

$32 + 16 + 4 + 1 = 53$

R/= $110101_2 = 53_{10}$

Operaciones binarias



• Suma Reglas básicas

- $0 + 0 = 0$
- $0 + 1 = 1$
- $1 + 0 = 1$
- $1 + 1 = 10$ --> en la suma 0 y va 1 de acarreo
- $1 + 1 + 1 = 1$ --> lleva 1

Norma

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Operaciones binarias

Resta reglas básicas

- 0 - 0 = 0
- 1 - 1 = 0
- 1 - 0 = 1
- 0 - 1 = 1 y acarreo 1

4b) 634 (octal)

6 3 4
8^2 8^1 8^0
64 8 1

6 × 64 = 384
3 × 8 = 24
4 × 1 = 4

384 + 24 + 4 = 412

$634_8 = 412_{10}$

4c) A A A (hexadecimal)

A A A
10 10 10
16^2 16^1 16^0
256 16 1

10 × 256 = 2560
10 × 16 = 160
10 × 1 = 10

2560 + 160 + 10

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$A\ A\ A_{16} = 2730_{10}$

4d) 1000001_2

1	0	0	0	0	0	1
2^6	2^5	2^4	2^3	2^2	2^1	2^0
64	32	16	8	4	2	1

$1 \times 64 = 64$
$0 \times 32 = 0$
$0 \times 16 = 0$
$0 \times 8 = 0$
$0 \times 4 = 0$
$0 \times 2 = 0$
$1 \times 1 = 1$

$1000001 = 65_{10}$

4e) 3765 (octal)

3	7	6	5
8^3	8^2	8^1	8^0
	64	8	1

$3 \times 512 = 1536$
$7 \times 64 = 448$
$6 \times 8 = 48$
$5 \times 1 = 5$

$1536 + 448 + 48 + 5 =$

$3765_8 = 2037_{10}$

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4f) f000c (hexadecimal)

f	0	0	0	c
15	0	0	0	12
16^4	16^3	16^2	16^1	16^0
65536	0	0	0	1

15 x 65536 = 983040
0 x 0 = 0
0 x 0 = 0
0 x 0 = 0
12 x 1 = 12

$983040 + 12 = 983052_{10}$

4g) 11111_2 (binario)

1	1	1	1	1
2^4	2^3	2^2	2^1	2^0
16	8	4	2	1

16 x 1 = 16
1 x 8 = 8
1 x 4 = 4
1 x 2 = 2
1 x 1 = 1

16 + 8 + 4 + 2 + 1 = 31

$11111_2 = 31_{10}$

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4h) 1 1 1 1 1 (Octa)

1 1 1 1 1
8^4 8^3 8^2 8^1 8^0
4096 512 64 8 1

$4096 \times 1 = 4096$
$512 \times 1 = 512$
$64 \times 1 = 64$
$8 \times 1 = 8$
$1 \times 1 = 1$

$4096 + 512 + 64 + 8 + 1 = 4681$

$11111_8 = 4681$

4i) 1 1 1 1 (Hexadecimal)

1 1 1 1
16^3 16^2 16^1 16^0
4096 256 16 1

$4096 \times 1 = 4096$
$256 \times 1 = 256$
$16 \times 1 = 16$
$1 \times 1 = 1$

$4096 + 256 + 16 + 1 = 4369$

$1111_{16} = 4369_{10}$

5) Convertir de binario a hexadecimal

5a) 1010101

5b) 100111101010 1

5a) 1 0 1 0 1 0 1 $_2$

0 1 0 1 0 1 0 1

5 5

55_{16}

0000 = 0
0001 = 1
0010 = 2
0011 = 3
0100 = 4
0101 = 5
0110 = 6
0111 = 7
1000 = 8
1001 = 9
1010 = A (10)
1011 = B (11)
1100 = C (12)
1101 = D (13)
1110 = E (14)
1111 = F (15)

5b) 1 0 0 1 1 1 1 0 1 0 1 0 1

0001 0011 1101 0101

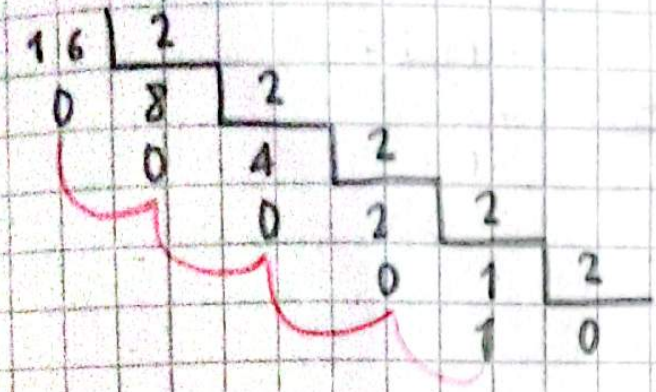
1 3 D 5

$13D5_{16}$

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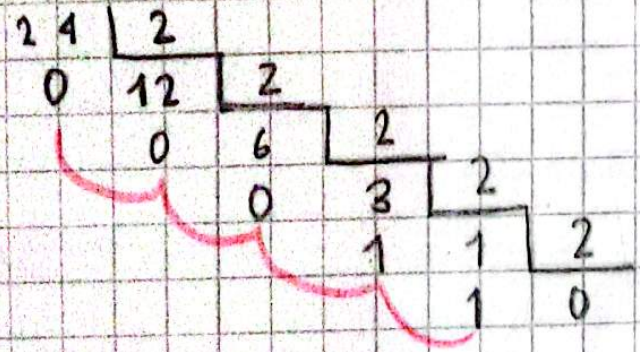
6) Sumar en binario y realizar la prueba

60) 16 + 24



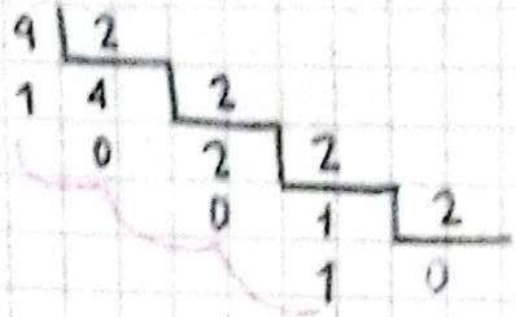
10000

$$\begin{array}{r} 10000 + \\ 11000 \\ \hline 101000_2 \end{array}$$

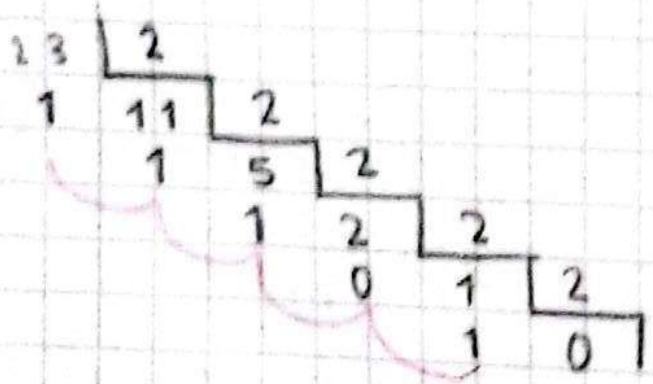


11000

6b) 9 + 23



1001



10111

```
 1 1 1 1
 1 0 1 1 1 +
     1 0 0 1
-----------
 1 0 0 0 0 0
```

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Prueba de verificación

6a) 16 + 24

16)

1	0	0	0	0
2^4	2^3	2^2	2^1	2^0
16	8	4	2	1
16	0	0	0	0

16 x 1 = 16 ✓

24)

1	1	0	0	0
2^4	2^3	2^2	2^1	2^0
16	8	4	2	1
16	8	0	0	0

16 x 1 = 16
8 x 1 = 8

16 + 8 = 24 ✓

6b) 9 + 23

9)

1	0	0	1
2^3	2^2	2^1	2^0
8	4	2	1

1 x 8 = 8
1 x 1 = 1

8 + 1 = 9 ✓

Norma

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23)

1	0	1	1	1
2^4	2^3	2^2	2^1	2^0
16	8	4	2	1

$16 \times 1 = 16$
$4 \times 1 = 4$
$2 \times 1 = 2$
$1 \times 1 = 1$

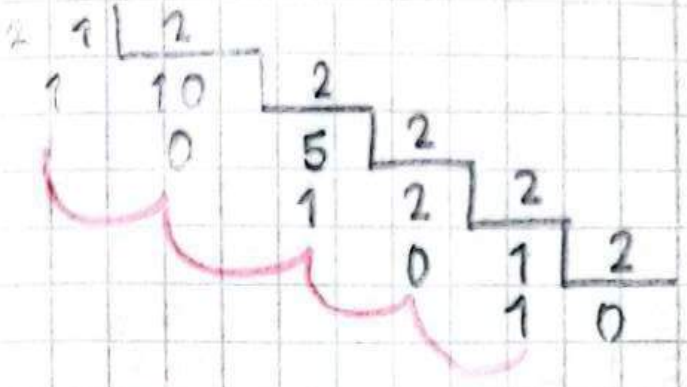
$16 + 4 + 2 + 1 = 23$ ✓

7) Restar en binario y realizar la prueba

7a) 37 - 21

37 | 2					
1	18 | 2				
	0	9 | 2			
		1	4 | 2		
			0	2 | 2	
				0	1 | 2
					1	0

1 0 0 1 0 1



1 0 1 0 1

Resta

$$\begin{array}{r} {\scriptstyle 1\ \ 1\ \ \ \ \ \ \ \ } \\ 1_0\,0\ 0\ 1\ 0\ 1 \\ -\ \ 1\ 0\ 1\ 0\ 1 \\ \hline 0\,1\,0\,0\,0\,0_2 \end{array}$$

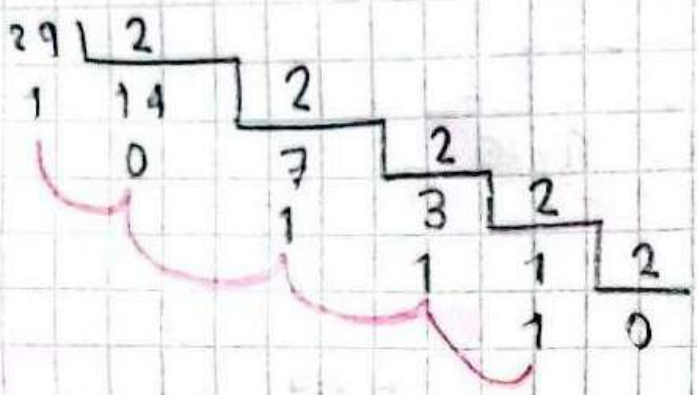
Nota

Para recordar las reglas básicas de resta en binario necesitamos saber:

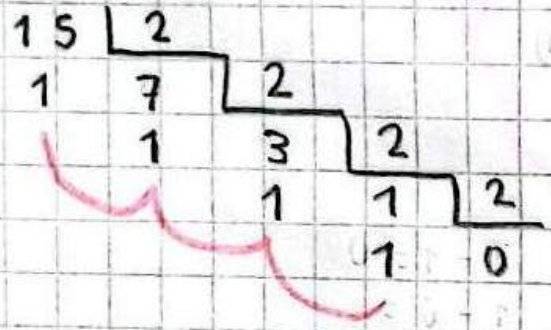
$0 - 0 = 0$
$1 - 1 = 0$
$1 - 0 = 1$
$0 - 1 = 1$ y acarreo 1

$1 - 1 = 0$
$1 - 0 = 1$
$0 - 0 = 0$
$0 - 1 = 1 \rightarrow$ acarreo

7b) 29 - 15



1 1 1 0 1



1 1 1 1

$$\begin{array}{r} {\scriptstyle 1\ \ 1\ \ 1\ \ 1} \\ 1\ 1\ 1\ 0\ 1 \\ -\ 1\ 1\ 1\ 1 \\ \hline 0\ 1\ 1\ 1\ 0_2 \end{array}$$

Prueba de verificación

7a) 37 - 21

37) 1 0 0 1 0 1
2^5 2^4 2^3 2^2 2^1 2^0
32 16 8 4 2 1

$32 \times 1 = 32$
$1 \times 4 = 4$
$1 \times 1 = 1$

$32 + 4 + 1 = 37$ ✓

21) 1 0 1 0 1
2^4 2^3 2^2 2^1 2^0
16 8 4 2 1

$1 \times 16 = 16$
$1 \times 4 = 4$
$1 \times 1 = 1$

$16 + 4 + 1 = 21$ ✓

7b) 29 - 15

29) 1 1 1 0 1
2^4 2^3 2^2 2^1 2^0
16 8 4 2 1

$16 \times 1 = 16$
$8 \times 1 = 8$
$1 \times 4 = 4$
$1 \times 1 = 1$

$16 + 8 + 4 + 1 = 29$ ✓

15) 1 1 1 1
2^3 2^2 2^1 2^0
8 4 2 1

$8 + 4 + 2 + 1 = 15$ ✓

Norma

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